

Supplementary Material: Zero-Incoherence Capacity of Interactive Encoding Systems

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1 Full Lean Handle Ledger

This supplement provides the complete Lean handle ledger cited by the manuscript. It includes every handle identifier, declaration name, and source module path.

ID	Handle	ID	Handle
AFM1	FactMatroid.determinesFact_iff_mem_factSpan Ssot/FactMatroid.lean	AFM2	FactMatroid.factMatroid_indep_iff Ssot/FactMatroid.lean
AFM3	FactMatroid.basisFacts_card_eq_finrank Ssot/FactMatroid.lean	AFM4	FactMatroid.basisFacts_minimal Ssot/FactMatroid.lean
AFM5	FactMatroid.minimalDetermining_basis Ssot/FactMatroid.lean	AFM6	FactMatroid.factRankFinset_le_card Ssot/FactMatroid.lean
AFM7	FactMatroid.coordProjection_range_le_card Ssot/FactMatroid.lean	AFM8	FactMatroid.factRankFinset_eq_card_of_indepFacts Ssot/FactMatroid.lean
AFM9	FactMatroid.factRankFinset_eq_total_of_ determinesAllFinset Ssot/FactMatroid.lean	AFM10	FactMatroid.factSpanFinset_eq_range_coordProjection_ dualMap Ssot/FactMatroid.lean
AFM11	FactMatroid.factRankFinset_eq_finrank_range_ coordProjection Ssot/FactMatroid.lean	AFM12	FactMatroid.factRankFinset_le_finrank_directionSpace Ssot/FactMatroid.lean
AFM13	FactMatroid.coordProjection_range_eq_card_of_ indepFacts Ssot/FactMatroid.lean	AFM14	FactMatroid.coordProjection_range_eq_total_of_ determinesAllFinset Ssot/FactMatroid.lean
BND1	ssot_upper_bound Ssot/Bounds.lean	BND2	non_ssot_lower_bound Ssot/Bounds.lean
BND3	ssot_advantage_unbounded Ssot/Bounds.lean	CAP2	ssot_guarantees_coherence Ssot/Coherence.lean
CAP3	non_ssot_permits_incoherence Ssot/Coherence.lean	CC1	CC.derivation_preserves_coherence_core
CIA1	ClaimClosure.side_information_requirement Ssot/ClaimClosure.lean	CIA3	ClaimClosure.finite_counting_converse Ssot/ClaimClosure.lean
CIA4	ClaimClosure.info_dof_counting_contradiction Ssot/ClaimClosure.lean	COH1	dof_one_implies_coherent Ssot/Coherence.lean
COH2	dof_gt_one_incoherence_possible Ssot/Coherence.lean	DER2	ClaimClosure.derivation_preserves_coherence_core Ssot/ClaimClosure.lean
FM6	FM.factRankFinset_eq_card_of_indepFacts	IND1	both_requirements_independent Ssot/Requirements.lean
IND2	both_requirements_independent Ssot/Requirements.lean	MFT3	MultiFact.exactRecovery_iff_properColoring Ssot/MultiFact.lean
MFT4	MultiFact.exists_exactRecovery_iff_colorable Ssot/MultiFact.lean	MFT5	MultiFact.binaryViews_nonclique_witness Ssot/MultiFact.lean
MFT6	MultiFact.binaryViews_colorable_two Ssot/MultiFact.lean	MFT7	MultiFact.binaryViews_not_colorable_one Ssot/MultiFact.lean
MFT8	MultiFact.binaryViews_oneTag_uniform_successProb_le_ half Ssot/MultiFact.lean	MFT9	MultiFact.exists_exactOn_iff_colorableOn Ssot/MultiFact.lean
MFT10	MultiFact.binaryViews_oneTag_exactOn_card_le_two Ssot/MultiFact.lean	MFT11	MultiFact.pairProperColoring_of_componentColorings Ssot/MultiFact.lean
MFT12	MultiFact.pairColorable_of_colorable Ssot/MultiFact.lean	MFT13	MultiFact.pair_product_clique_budget_lower_bound Ssot/MultiFact.lean
MFT14	MultiFact.exists_exactOn_card_eq_maxColorableCard Ssot/MultiFact.lean	MFT15	MultiFact.binaryViews_maxColorableCard_one_eq_two Ssot/MultiFact.lean
MFT16	MultiFact.pair_product_success_card_lower_bound Ssot/MultiFact.lean	MFT17	MultiFact.pair_product_independent_lower_bound Ssot/MultiFact.lean
MFT18	MultiFact.exactOn_mass_le_maxColorableMass Ssot/MultiFact.lean	MFT19	MultiFact.exists_exactOn_mass_eq_maxColorableMass Ssot/MultiFact.lean
MFT20	MultiFact.colorable_iff_graphColorable Ssot/MultiFact.lean	MFT21	MultiFact.block_power_independent_lower_bound Ssot/MultiFact.lean
MFT22	MultiFact.block_add_independent_lower_bound Ssot/MultiFact.lean	MFT23	MultiFact.exactRateDistortionValue_upper Ssot/MultiFact.lean
MFT24	MultiFact.exists_exactRateDistortionValue Ssot/MultiFact.lean	MFT25	MultiFact.pairConfusable_iff_strongProdAdj Ssot/MultiFact.lean
MFT26	MultiFact.pairConfusabilityGraph_eq_strongProd Ssot/MultiFact.lean	MFT29	MultiFact.concatBlockConfusable_iff_strongProdAdj Ssot/MultiFact.lean
MFT30	MultiFact.blockConfusabilityGraph_addIsoStrongProd Ssot/MultiFact.lean	MFT31	MultiFact.blockMaxIndependentCard_eq_ graphMaxIndependentCard Ssot/MultiFact.lean
MFT32	MultiFact.shannonLowerCapacity_eq_iSup_graphRates Ssot/MultiFact.lean	MFT33	MultiFact.blockRate_le_blockRate_mul Ssot/MultiFact.lean
MFT34	MultiFact.blockConfusable_iff_strongPowAdj Ssot/MultiFact.lean	MFT35	MultiFact.blockConfusabilityGraph_eq_strongPow Ssot/MultiFact.lean
MFT36	MultiFact.shannonLowerCapacity_eq_ graphShannonLowerCapacity Ssot/MultiFact.lean	MFT38	MultiFact.strongPow_addIsoStrongProd Ssot/MultiFact.lean

ID	Handle	ID	Handle
MFT39	MultiFact.graphMaxIndependentCard_strongPow_add_eq_strongProd Ssot/MultiFact.lean	MFT40	MultiFact.graphMaxIndependentCard_strongProd_lower_bound Ssot/MultiFact.lean
MFT41	MultiFact.graphMaxIndependentCard_strongPow_mul_lower_bound Ssot/MultiFact.lean	MFT42	MultiFact.graphPowerRate_le_graphPowerRate_mul Ssot/MultiFact.lean
MFT43	MultiFact.graphNegLogSeq_subadditive Ssot/MultiFact.lean	MFT44	MultiFact.tendsto_graphPowerRateNat_graphShannonCapacityReal Ssot/MultiFact.lean
MFT45	MultiFact.graphPowerRate_le_graphShannonCapacityReal Ssot/MultiFact.lean	MFT46	MultiFact.graphShannonCapacityReal_nonneg Ssot/MultiFact.lean
MFT47	MultiFact.iSup_graphPowerRate_eq_graphShannonCapacityReal Ssot/MultiFact.lean	MFT48	MultiFact.shannonCapacityReal_eq_iSup_blockRate Ssot/MultiFact.lean
MFT49	MultiFact.blockRate_le_shannonCapacityReal Ssot/MultiFact.lean	MFT50	MultiFact.compl_strongPow_adj_iff_exists Ssot/MultiFact.lean
MFT51	MultiFact.complStrongPow_colorable Ssot/MultiFact.lean	MFT52	MultiFact.graphMaxIndependentCard_strongPow_le_complChromaticPow Ssot/MultiFact.lean
MFT53	MultiFact.graphPowerRate_le_log_complChromatic Ssot/MultiFact.lean	MFT54	MultiFact.graphShannonCapacityReal_le_log_complChromatic Ssot/MultiFact.lean
MFT55	MultiFact.graphShannonCapacityReal_le_log_complChromaticNumber Ssot/MultiFact.lean	MFT56	MultiFact.graphShannonLowerCapacity_eq_ofReal_graphShannonCapacityReal Ssot/MultiFact.lean
MFT57	MultiFact.graphThetaPSDLogUpper_eq_graphThetaLogUpper Ssot/MultiFact.lean	MFT58	MultiFact.graphShannonCapacityReal_le_graphThetaPSDLogUpper Ssot/MultiFact.lean
MFT59	MultiFact.graphThetaPSDLogUpper_strongProd_le Ssot/MultiFact.lean	MFT60	MultiFact.graphThetaPSDLogSeq_subadditive Ssot/MultiFact.lean
MFT61	MultiFact.tendsto_graphThetaPSDPowerRateNat_graphThetaPSDAsymptoticUpper Ssot/MultiFact.lean	MFT62	MultiFact.graphThetaPSDAsymptoticUpper_le_graphThetaPSDLogUpper Ssot/MultiFact.lean
MFT63	MultiFact.lInf_graphThetaPSDPowerRate_eq_graphThetaPSDAsymptoticUpper Ssot/MultiFact.lean	MFT64	MultiFact.lovaszOrthoLogUpper_eq_graphThetaLogUpper Ssot/MultiFact.lean
MFT65	MultiFact.lovaszThetaPrimalLogUpper_eq_graphThetaPSDLogUpper Ssot/MultiFact.lean	MFT66	MultiFact.graphThetaLiftedDualLogUpper_eq_graphThetaSchurDualLogUpper Ssot/MultiFact.lean
MFT67	MultiFact.graphOneShotLogMaxIndependentCard_le_graphThetaLogUpper Ssot/MultiFact.lean	MFT68	MultiFact.graphPowerRate_le_graphThetaPSDPowerRate Ssot/MultiFact.lean
MFT69	MultiFact.graphShannonCapacityReal_le_graphThetaPSDAsymptoticUpper Ssot/MultiFact.lean	MFT70	MultiFact.lovaszThetaPrimalAsymptoticUpper_eq_graphThetaPSDAsymptoticUpper Ssot/MultiFact.lean
MFT71	MultiFact.graphShannonCapacityReal_le_lovaszThetaPrimalAsymptoticUpper Ssot/MultiFact.lean	MFT72	MultiFact.graphThetaSchurDualLogUpper_strongProd_le Ssot/MultiFact.lean
MFT73	MultiFact.graphThetaSchurDualLogUpper_iso_eq Ssot/MultiFact.lean	MFT74	MultiFact.graphThetaSchurDualLogSeq_subadditive Ssot/MultiFact.lean
MFT75	MultiFact.tendsto_graphThetaSchurDualPowerRateNat_graphThetaSchurDualAsymptoticUpper Ssot/MultiFact.lean	MFT76	MultiFact.lInf_graphThetaSchurDualPowerRate_eq_graphThetaSchurDualAsymptoticUpper Ssot/MultiFact.lean
MFT77	MultiFact.graphThetaSchurDualAsymptoticUpper_le_graphThetaPSDAsymptoticUpper Ssot/MultiFact.lean	MFT78	MultiFact.graphThetaSchurDualAsymptoticUpper_le_lovaszThetaPrimalAsymptoticUpper Ssot/MultiFact.lean
MFT79	MultiFact.lovaszThetaDualLogUpper_eq_graphThetaSchurDualLogUpper Ssot/MultiFact.lean	MFT80	MultiFact.lovaszThetaDualAsymptoticUpper_eq_graphThetaSchurDualAsymptoticUpper Ssot/MultiFact.lean
MFT81	MultiFact.lovaszThetaDualAsymptoticUpper_le_lovaszThetaPrimalAsymptoticUpper Ssot/MultiFact.lean	MFT82	MultiFact.lovaszThetaDualLogUpper_eq_lovaszThetaPrimalLogUpper Ssot/MultiFact.lean
MFT83	MultiFact.lovaszThetaDualAsymptoticUpper_eq_lovaszThetaPrimalAsymptoticUpper Ssot/MultiFact.lean	MFT84	MultiFact.graphShannonCapacityReal_le_lovaszThetaAsymptoticUpper Ssot/MultiFact.lean
MFT85	MultiFact.graphShannonCapacityReal_eq_lovaszThetaAsymptoticUpper_clusterGraph Ssot/MultiFact.lean	MFT86	MultiFact.graphShannonCapacityReal_eq_log_card_clusterGraph Ssot/MultiFact.lean
MFT87	MultiFact.lovaszThetaAsymptoticUpper_eq_log_card_clusterGraph Ssot/MultiFact.lean	MFT88	MultiFact.confusabilityGraph_eq_clusterGraph_transcriptLabel Ssot/MultiFact.lean
MFT89	MultiFact.shannonCapacityReal_eq_shannonLovaszThetaAsymptoticUpper_of_fiberCoherent Ssot/MultiFact.lean	MFT90	MultiFact.shannonCapacityReal_eq_log_transcriptFiberCard_of_fiberCoherent Ssot/MultiFact.lean
MFT91	MultiFact.shannonLovaszThetaAsymptoticUpper_eq_log_transcriptFiberCard_of_fiberCoherent Ssot/MultiFact.lean	MFT96	MultiFact.shannonLovaszThetaAsymptoticUpper_eq_log_connectedComponents_of_confusableTransitive Ssot/MultiFact.lean
MFT100	MultiFact.shannonCapacityReal_eq_shannonLovaszThetaAsymptoticUpper_of_meetWitnessed Ssot/MultiFact.lean	MFT101	MultiFact.shannonCapacityReal_eq_log_connectedComponents_of_meetWitnessed Ssot/MultiFact.lean
MFT102	MultiFact.shannonLovaszThetaAsymptoticUpper_eq_log_connectedComponents_of_meetWitnessed Ssot/MultiFact.lean	MFT107	MultiFact.ViewCompositionClosed Ssot/MultiFact.lean
MFT108	MultiFact.confusableTransitive_iff_viewCompositionClosed Ssot/MultiFact.lean	MFT109	MultiFact.confusable_iff_exists_viewClusterAdj Ssot/MultiFact.lean
MFT110	MultiFact.SupportConfusable Ssot/MultiFact.lean	MFT111	MultiFact.deterministicSupportConfusable_iff_confusable Ssot/MultiFact.lean
MFT112	MultiFact.deterministicSupportConfusabilityGraph_eq_confusabilityGraph Ssot/MultiFact.lean	MFT113	MultiFact.card_state_eq Ssot/MultiFact.lean
MFT114	MultiFact.confusableOracle_eq_true_iff Ssot/MultiFact.lean	MFT115	MultiFact.confusableOrderedPairs Ssot/MultiFact.lean
MFT116	MultiFact.card_state_prod_eq Ssot/MultiFact.lean	MFT117	MultiFact.card_confusableOrderedPairs_le_naive_bound Ssot/MultiFact.lean
MFT118	MultiFact.observe_eq_iff_subset_agreeSet Ssot/MultiFact.lean	MFT119	MultiFact.confusable_iff_exists_view_subset_agreeSet Ssot/MultiFact.lean
MFT120	MultiFact.agreeSet_statePerm_eq Ssot/MultiFact.lean	MFT121	MultiFact.confusable_statePerm_iff Ssot/MultiFact.lean
MFT122	MultiFact.confusable_congr_agreeSet Ssot/MultiFact.lean	MFT123	MultiFact.confusabilityGraph_statePermIso Ssot/MultiFact.lean

ID	Handle	ID	Handle
MFT124	MultiFact.exists_confusabilityGraph_iso_send Ssot/MultiFact.lean	MFT125	MultiFact.agreementUpwardFamily_upwardClosed Ssot/MultiFact.lean
MFT126	MultiFact.confusable_iff_agreeSet_mem_ agreementUpwardFamily Ssot/MultiFact.lean	MFT127	MultiFact.confusable_viewFamilyOfAgreementUpwardFamily_ iff Ssot/MultiFact.lean
MFT128	MultiFact.exists_viewFamily_iff_agreementClassified Ssot/MultiFact.lean	MFT129	MultiFact.agreeSet_eq_univ_iff Ssot/MultiFact.lean
MFT130	MultiFact.exists_distinct_states_with_agreeSet_eq_iff Ssot/MultiFact.lean	MFT131	MultiFact.upwardClosed_family_eq_on_proper_subsets_ of_same_relation Ssot/MultiFact.lean
MFT132	MultiFact.agreementOracle_eq_true_iff Ssot/MultiFact.lean	MFT133	MultiFact.agreementOracle_eq_confusableOracle Ssot/MultiFact.lean
MFT134	MultiFact.agreementOracleViewWork_le Ssot/MultiFact.lean	MFT135	MultiFact.agreementOracleTotalWork_le Ssot/MultiFact.lean
MFT136	MultiFact.meetWitnessed_implies_transitive_confusability Ssot/MultiFact.lean	MFT137	MultiFact.transitive_confusability_implies_cluster_graph Ssot/MultiFact.lean
OBS1	ObserverModel.exact_recovery_implies_pair_injective paper1/Paper1IT/ObserverTagModel.lean	OBS2	ObserverModel.pair_injective_implies_exact_recovery paper1/Paper1IT/ObserverTagModel.lean
OBS3	ObserverModel.fiber_card_le_tag_alphabet paper1/Paper1IT/ObserverTagModel.lean	OBS4	ObserverModel.exact_recovery_global_count paper1/Paper1IT/ObserverTagModel.lean
OBS5	ObserverModel.clique_card_le_tag_alphabet paper1/Paper1IT/ObserverTagModel.lean	OM1	OM.DeterministicObservable.entropy_coarsen_le_ sourceEntropy
PRB47	ObserverModel.sourceEntropy_eq_ observationTagEntropy_add_conditionalEntropyGivenPair paper1/Paper1IT/ProbabilisticFinite.lean	PRB55	ObserverModel.successSet_clique_entropy_le_log_tags paper1/Paper1IT/FanoFinite.lean
PRB68	ObserverModel.deterministicKernel_entropy_data_ processing paper1/Paper1IT/ProbabilisticFinite.lean	PRB73	ObserverModel.observationTagEntropy_gap_nonneg paper1/Paper1IT/ProbabilisticFinite.lean
PRB74	ObserverModel.observationTagEntropy_gap_eq_zero_of_ klDiv_zero_uniform paper1/Paper1IT/ProbabilisticFinite.lean	PRB81	ObserverModel.decodedOutputEntropy_le_ mutualInfoDeterministic paper1/Paper1IT/ProbabilisticFinite.lean
PRB82	ObserverModel.decodedOutputEntropy_source_gap_ nonneg paper1/Paper1IT/ProbabilisticFinite.lean	PRB83	ObserverModel.decodedOutputEntropy_log_gap_nonneg paper1/Paper1IT/ProbabilisticFinite.lean
PRB85	ObserverModel.decodedOutputEntropy_fano_budgeted paper1/Paper1IT/FanoFinite.lean	PRB87	ObserverModel.decodedOutputEntropy_fano_observation_ only paper1/Paper1IT/FanoFinite.lean
PRB90	ObserverModel.DeterministicObservable.entropy_coarsen_ le_sourceEntropy paper1/Paper1IT/ProbabilisticFinite.lean	PRB93	ObserverModel.decodedOutputEntropy_gap_eq_zero_of_ klDiv_zero_uniform paper1/Paper1IT/ProbabilisticFinite.lean
PRB94	ObserverModel.decodedOutputEntropy_gap_pos_implies_ klDiv_ne_zero paper1/Paper1IT/ProbabilisticFinite.lean	PRB95	ObserverModel.decodedOutputObservable_entropy_le_ sourceEntropy paper1/Paper1IT/ProbabilisticFinite.lean
RAT1	Ssot/ClaimClosure.lean ClaimClosure.rate_incoherence_step	RED1	Ssot/ClaimClosure.lean ClaimClosure.redundancy_incoherence_equiv
REQ1	structural_facts_fixed_at_definition Ssot/Foundations.lean	REQ2	Ssot/ClaimClosure.lean definition_hooks_necessary Ssot/Requirements.lean
REQ3	introspection_necessary_for_verification Ssot/Requirements.lean	SOT1	Ssot/Completeness.lean
TRI1	timing_trichotomy_exhaustive Ssot/Foundations.lean		

2 Mechanized Verification Details

The main theorem chains of this work are machine-checked in Lean 4, and the cited proof modules compile with zero `sorry` placeholders. The artifact covers the following theorem packages:

- **Threshold and Unified-Converse:** Coherence, independent rate, the side-information lower bound, pair injectivity, clique/fiber/global budget bounds, conditional-entropy and decoder-output reformulations, and deterministic data processing.
- **Graph and Asymptotic-Capacity:** Explicit confusability graph objects, exact recovery as colorability, strong-power identification for block confusability, Fekete-style asymptotic Shannon capacity, and a fixed Lovász- ϑ upper theory.
- **Equality and SSOT:** The clique-fiber equality theorem, meet-witness sufficient conditions, the fiber-coherence collapse criterion, and the realizability iff theorem.
- **Affine Fact-Matroid:** Semantic determination as span membership of coordinate functionals and the identification of minimal determining fact sets with matroid bases.

3 Verification Statistics

Metric	Local (Ssot/)	Total Development
Source Files	32	85
Lines of Code	11903	25743
Theorems/Lemmas	557	1234
Placeholders (<code>sorry</code>)	0	0

Table 2: Summary of mechanized verification statistics for the Paper 2 artifact.

4 Claim Mapping to Lean Handles

This section provides the direct mapping between paper handles (Theorem, Corollary, Lemma) and the mechanized proof handles in the Lean 4 artifact.

Claim Mapping to Lean Handles

Paper handle	Status	Lean support
cor:integrity-threshold	Full	COH2, COH1
cor:	Full	AFM3, FM6, AFM7, AFM8, AFM9, AFM10, AFM11
matroid-capacity-bounds		
lem:confusability-clique	Full	OBS2
lem:info-dof	Full	CIA4
thm:affine-fact-matroid	Full	AFM1, AFM2, AFM4, AFM5, AFM12
thm:asymptotic-capacity	Full	MFT7, MFT13, MFT14, MFT15, MFT16, MFT17, MFT22, MFT23
thm:causal-necessary	Full	REQ2
thm:coherence-capacity	Full	CAP3, CAP2
lem:	Full	OBS2
confusability-converse		
lem:derivation-excludes	Full	CC1
lem:	Full	COH2
dof-gt-one-incoherence		
lem:dof-one-coherence	Full	COH1
lem:	Full	OBS1, OBS2, OBS5, OM1
equivalence-viewpoint		
lem:fano-converse	Full	CIA1
lem:	Full	OBS3, OBS4
finite-error-budgeted		
lem:independence	Full	IND1, IND2
lem:lower-bound	Full	BND2
lem:	Full	MFT8, MFT9, MFT12
multifact-colorability		
lem:	Full	MFT10
multifact-max-success		
lem:multifact-nonclique	Full	MFT3, MFT4, MFT5, MFT6
lem:multifact-product	Full	MFT18, MFT19, MFT20, MFT21
lem:	Full	MFT11
multifact-success-set		
lem:pair-injective	Full	OBS1
lem:provenance-necessary	Full	REQ3
lem:side-info	Full	CIA1
lem:ssot-iff	Full	SOT1
lem:timing-forces	Full	REQ1, TRI1
lem:unbounded-gap	Full	BND3
lem:upper-bound	Full	BND1

Auto summary: mapped 29/29 (full=29, derived=0, unmapped=0).